

RESEARCH INTERESTS

Autonomous agents and multi-agent interaction, with a focus on how structured representations and constraints shape agent behavior. I am particularly interested in bridging agent architectures with human–AI interaction and world models, and in making agent-driven systems more interpretable, controllable, and trustworthy. My broader interests include safety-oriented AI design, especially in settings involving long-term adaptation and sustained agent–user interaction.

SELECTED RESEARCH PROJECTS

Emotion Landscape — Interactive Storytelling System New York City, USA
Columbia DSL (Digital Storytelling Lab) 2025 - Present

- Investigated how structured prompting strategies affect emergent behavior in interactive, agent-driven narrative environments. Designed modular interaction logic to study coordination between narrative agents and user actions.
- Built an assets pipeline and explored affective design patterns for interactive spaces.

Prompt Markdown+ 2025 - Present
Independent Research

- Designed a structured prompt language to improve controllability and interpretability, and to study how abstraction layers affect agent failure modes and user mental models in human–AI interaction.
- Built an early prototype and evaluation plan to measure agent consistency under constrained prompt schemas (preparing user study).

Interactive Engine (In.js) 2025 - Present
Independent Research

- Built a modular JavaScript-based interactive engine to experiment with agent-driven narrative logic.
- Exploring how architectural choices affect scalability and agent coordination in interactive systems.

RESEARCH EXPERIENCE

Human-Centered Emergency Preparedness Research, Columbia University
Research Assistant with Professor Harry West, Columbia Engineering, in collaboration with the NYC Preparedness & Recovery Institute (PRI) Summer 2026 – Present

- Develop product concepts, interaction frameworks, and iterative prototypes from research findings, focusing on rapid communication, community activation, and emergency decision support.
- Conduct cross-case synthesis across interviews, workshops, and stakeholder engagements with city agencies, community organizations, and local leaders, identifying recurring breakdowns in rapid consultation, information handoffs, implementation, and feedback under emergency conditions.
- Conduct interviews, workshops, and prototype testing with public-health, emergency-preparedness, and community stakeholders, including participants from NYC DOHMH, CDC, and other government and community organizations, examining how community and frontline knowledge can better inform time-sensitive institutional decision-making.

Interactive Asylum Case-Review System Summer 2026
Migration between France and North Africa, Professor Madeleine Dobie

- Designed an interactive asylum case-review system that placed users in the role of an adjudicator reviewing a fictional case file. The interface explicitly warned that the evidence was incomplete but still required a decision, with no option to abstain—turning evidentiary uncertainty and institutional obligation into an interaction rather than simply describing them. [[Project link](#)]

Digital Storytelling Lab (DSL), Columbia University 2025 - Present

- Participated in project-based research on interactive systems and computational storytelling, with an emphasis on iterative prototyping and analysis of agent behavior, user interaction, and narrative dynamics.
- Selected demos: [[YouTube link](#)]

Participant

- Participated in rapid prototyping of interactive AI systems; responsibilities included prompt engineering, narrative design, and interface logic.

NSF(National Science Foundation) I-Corps

Princeton University

Propelus Certificate of Completion

Spring 2024

- Completed National Science Foundation (NSF) Innovation Corps (I-Corps™) Northeast Hub -funded training in research commercialization, customer discovery, and early-stage product validation.
- Received micro-funding to identify initial customers and define a prototype concept, collaborating with entrepreneurs and mentors across NSF's innovation network.
- Designed an AI system informed by Buddhist contemplative principles to support reflective user interaction. Secured NSF I-Corps funding and conducted over 100 user interviews to validate needs among monasteries, practitioners, and scholars.

EDUCATION

Columbia University

New York City, USA

B.A. in Computer Science & Cognitive Science

- GPA 4.0 / 4.0

2025 - Present

Chinese Academy of Social Sciences (CASS)

Beijing, China

M.Phil., Humanities (textual and visual analysis; narrative structures)

- Overall GPA: 3.70/4.0

Beijing University of Technology (BJUT)

Beijing, China

Bachelor of Engineering, Civil Engineering

ACADEMIC HONORS & AWARDS

Charlotte W. Newcombe Foundation Endowed Scholarship, Columbia University

2026 - 27

- Selected for a named endowed scholarship reserved for the School's most talented current students.

Millennium Fellow, Millennium Campus Network & United Nations Academic Impact

2026 - 27

- Selected for the U.S. & MENA Region Virtual Exchange Program.

Global Learning Scholarship, Columbia University

Summer 2026

Dean's List, Columbia University School of General Studies

All eligible terms

Exceptional Master's Thesis, Chinese Academy of Social Sciences

2011

SELECTED PRODUCT & TECHNOLOGY EXPERIENCE

- Developed product and platform experience across fintech, digital publishing, blockchain infrastructure, and AI prototypes, with a focus on turning complex technical ideas into usable products.
- Worked on web, iOS, and Android product workflows, connecting technical feasibility with interface logic, user experience, and implementation priorities.
- Translated financial platforms, publishing workflows, and AI interaction concepts into structured, testable prototypes.
- Prior experience in venture-style project development, cross-cultural coordination, and turning early concepts into clear public-facing products and narratives.

TECHNICAL SKILLS

- Programming: Python, Java, C/C++, JavaScript
- Systems/Dev: Linux, Git, CLI
- AI/Agents: Multi-agent Prompting, HCI Prototyping
- Prototyping & Media: Interactive demos, Video-based documentation, Godot